

PulseInsightAI – Project Design

Project Title:

PulseInsightAI – Multi-Disease Risk Prediction Using ML

① Objective

To build a machine learning system that predicts the risk of diabetes, heart disease, and stroke using real-world public health survey data (CDC BRFSS) and provides actionable insights for healthcare and preventive measures.

② Dataset

CDC BRFSS Chronic Disease Survey (latest year)

Rows: ~400,000

Features: ~50+ (demographics, lifestyle, health indicators)

Target variables:

Diabetes Risk → 0/1

Heart Disease Risk → 0/1

Stroke Risk → 0/1

Multi-label classification problem: each patient can have 0, 1, 2, or 3 diseases.

③ Input Features

Demographics:

Age, Gender, Race, Education, Income, Region

Lifestyle & Habits:

Smoking, Alcohol use, Exercise frequency, Diet, Sleep

Health Measurements:

BMI, Blood pressure, Cholesterol, Past conditions, Medications

Other Factors:

Family history, Stress / mental health indicators (if available)

④ Target Variables

Diabetes → 0 / 1

Heart Disease → 0 / 1

Stroke → 0 / 1

Multi-label binary classification problem.

⑤ Data Cleaning & Preprocessing

Handle missing values (drop or impute)

Encode categorical features (one-hot or label encoding)

Scale numeric features (StandardScaler / MinMaxScaler)

Feature engineering:

BMI categories

Lifestyle composite scores

Age groups

⑥ Exploratory Data Analysis (EDA)

Distribution of target variables

Correlation heatmap of features

Visualization of key risk factors

Check for imbalanced classes

⑦ Modeling

Step 1 – Baseline:

Multi-output Logistic Regression (for each disease)

Step 2 – Advanced Models:

Random Forest (multi-output)

Gradient Boosting / XGBoost (multi-output)

Step 3 – Model Evaluation:

Per disease: Accuracy, F1-score, Precision, Recall, ROC-AUC

Multi-label metrics: Hamming Loss, Subset Accuracy

Confusion matrix per disease

8 Dimensionality Reduction & Clustering

PCA, t-SNE, or UMAP for 2D/3D visualization

DBSCAN or KMeans clustering to identify:

High-risk patient clusters

Healthy / low-risk clusters

Visualize clusters to show multi-disease patterns

9 Feature Importance & Interpretability

Use tree-based model feature importance

SHAP / LIME for per-disease interpretability

Identify top risk factors for each disease

Visualize findings with bar plots / summary plots

10 Actionable Insights

Highlight high-risk patient groups

Recommend lifestyle adjustments or preventive measures

Identify combinations of factors leading to multiple diseases

Provide dashboards/plots for easy understanding

11 Portfolio / Presentation

Interactive dashboard with Plotly or Streamlit

Clear visualizations for clusters, risk distribution, feature importance

Documented Jupyter Notebook with all steps:

Introduction → EDA → Cleaning → Modeling → Clustering → Insights → Conclusion

README including:

Problem statement

Dataset description

Methods used

Key results & insights

How it could be applied in healthcare

12 Tools & Libraries

Python 3.x

Pandas, Numpy, Matplotlib, Seaborn

Scikit-learn, XGBoost, SHAP, UMAP-learn

Plotly / Streamlit for dashboard

Optional: Jupyter Notebook or VSCode

✓ Outcome

Multi-disease risk prediction system

Visual and actionable insights for healthcare professionals
Portfolio-worthy, recruiter-attractive ML project